

Classification Algorithms

ML algorithms divided into

- *Supervised algorithms*
 - *Regression*
 - *Linear regression*
 - *Ridge regression*
 - *Lasso Regression*
 - *Elastic net regression*
 - *Classification*
 - *Logistic Regression*
 - *Decision Tree*
 - *Naive Bayes*
 - *KNN*
 - *SVM*
 - *Ensemble methods*
 - *Bagging*
 - ✓ *Random Forest*
 - *Boosting*
 - ✓ *Ada boost*
 - ✓ *Gradient Boost*
 - ✓ *XG boost*
- *Unsupervised*
- *Reinforcement*

Classification

The data has both input data and output data,

output data type of categorical data

Example: Visa data set the Target column: case – status

- *Certified*
- *Denied*

Which means we are classifying the Observations

Example: Bank loan defualters : Col = Loan defualter

- *Yes*
- *No*

First example and second example the target column has only two classes

Bi – classification problem

Example: Based On your marks ,You want to divide the students

- *Topper*
- *Average*
- *Dull*

the target column has more than two classes

Multi – classification problem

Example: Google maps

The location might be correct or wrong

The Marker might be corect or wrong

Both location and Marker are wrong

Logistic Regression

Objective: I want to Know the subscription of magzine based on age

<i>Age</i>	<i>Subscription</i>
30	Yes
20	No
40	Yes
25	?

we knows only Linear regression

$$y = b_0 + b_1 * x_1$$

$$\text{Subscription} = b_0 + b_1 * \text{age} \text{ ===== (1)}$$

Is this equation is correct or not?

LHS = categorical data (expected Yes or No)

RHS = Numerical data

It is a wrong equation

<i>Age</i>	<i>Subscription</i>	<i>Subscription</i>
30	Yes	1
20	No	0
40	Yes	1
25	?	?

Next qn:

How can I make equation 1 ,Output to get 1's and 0's

Idea instead of giving y ===== $p(y) \geq 0.5$ === 1

$p(y) < 0.5$ ===== 0

$p(\text{Subscription}) = b_0 + b_1 * \text{age}$ ===== (2)

assume that $b_0 = 10$ and $b_1 = 20$

$p(\text{Subscription}) = 10 + 20 * \text{age}$ ===== (2)

$\text{age} = 30$ $p(\text{subscription}) = 10 + 20 * 30 = 610$

Age	Subscription	Subscription	$p(\text{Subscription})$ $= 10 + 20 * \text{age}$
30	Yes	1	10+20*30=610
20	No	0	
40	Yes	1	
25	?	?	

The equation (2) problem is

*We are expecting RHS : $b_0 + b_1 * \text{age}$ from 0 to 1*

but RHS varies from $-\infty$ to ∞

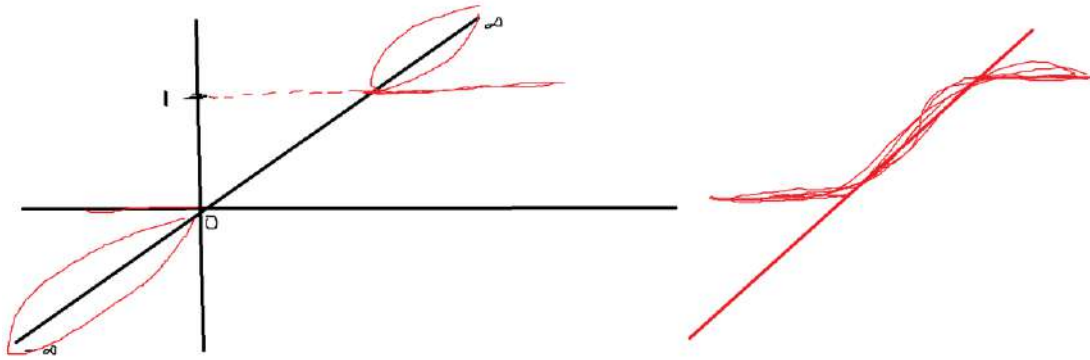
$\text{Subscription} = b_0 + b_1 * \text{age}$ ===== (1)

It is not a regression problem, It is a classification

Classification needs only yes (1) and No (0)

$p(\text{Subscription}) = b_0 + b_1 * \text{age}$ ===== (2)

probability ranges only 0 to 1, but my RHS is giving $-\infty$ to ∞



Idea is : RHS equation need to modify, such way that we need to get only 0 and 1

Rule – 1 : It should be postive value

Rule – 2: The value ranges only 0 to 1

Qn: which math equation will help us to get the answer

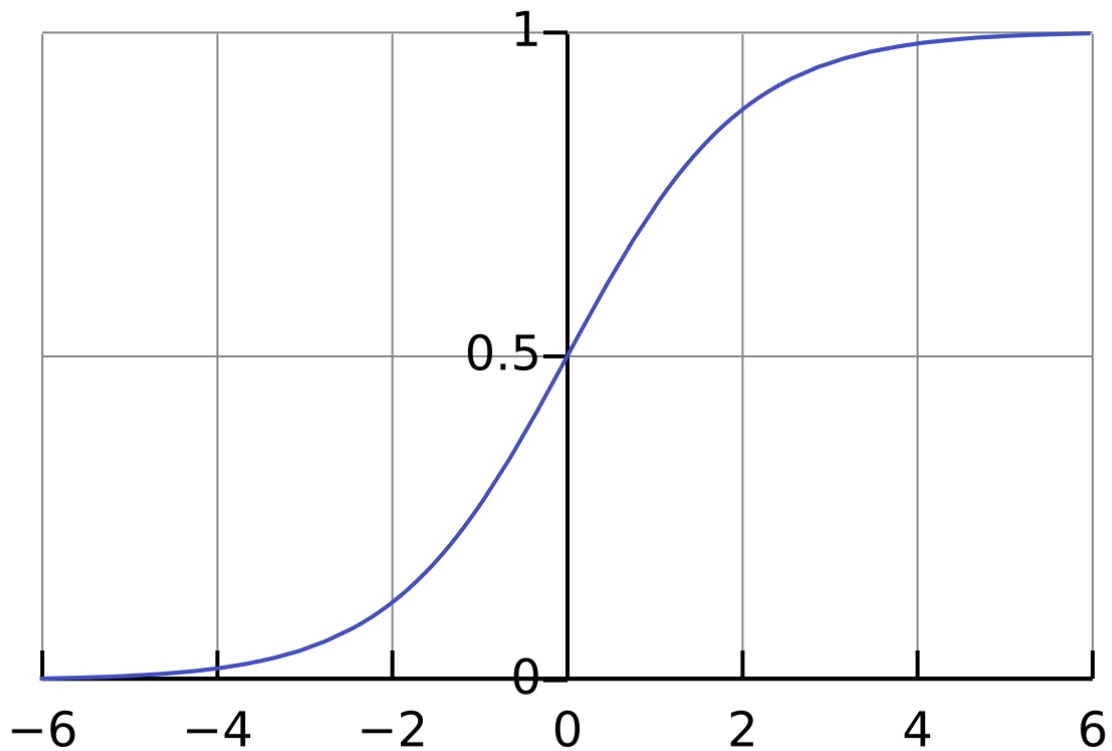
The answer is : Sigmoid Function

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma(y) = \frac{1}{1+e^{-y}}$$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

- *Sigmoid function value ranges from 0 to 1*
- *The curve look like 's' curve*



$$p(\text{Subscription}) = \sigma(b_o + b_1 * \text{age})$$

$$p(\text{Subscription}) = \frac{1}{1 + e^{-(b_o + b_1 * \text{age})}}$$

$$p(y) = \frac{1}{1 + e^{-(b_o + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n)}} \text{ ===== (3)}$$

we wil calculate b_1, b_2 ols method

we will substitute our data x_1 and x_2

$$\frac{1}{1 + e^{-(b_o + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n)}} \text{ === between 0 to 1}$$

$$\frac{1}{1 + e^{-(b_o + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n)}} = 0.6$$

$$p(y) = 0.6 \text{ ===== } > y = 1 \text{ yes}$$

$$\frac{1}{1 + e^{-(b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n)}} = 0.2$$

$$p(y) = 0.2 \implies y = 0 \text{ no}$$

Why sigmoid

- *sigmoid is also called as an Activation function*
- *Activation functions introduce Non linearity in the Neurons*
- *So the neuron able to understand the complex patterns easily*
- *we have sigmoid, softmax, Tanh, RelU, Leaky RelU*
- *will learn more in Deep learning*

$$\text{Subscription} = b_0 + b_1 * \text{age} \implies (1)$$

$$p(\text{Subscription}) = b_0 + b_1 * \text{age} \implies (2)$$

$$p(\text{Subscription}) = \frac{1}{1 + e^{-(b_0 + b_1 * \text{age})}} \implies (3)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\text{case - 1 : } x = -\infty$$

$$\sigma(-\infty) = \frac{1}{1 + e^{-(-\infty)}}$$

$$\sigma(-\infty) = \frac{1}{1 + e^{\infty}}$$

$$\sigma(-\infty) = \frac{1}{1 + \infty}$$

$$\sigma(-\infty) = \frac{1}{\infty} = 0$$

$$\text{case - 2 : } x = \infty$$

$$\sigma(\infty) = \frac{1}{1 + e^{-(\infty)}}$$

$$\sigma(\infty) = \frac{1}{1 + e^{-\infty}}$$

$$\sigma(\inf) = \frac{1}{1+0}$$

$$\sigma(\inf) = 1$$

case - 3 : $x = 0$

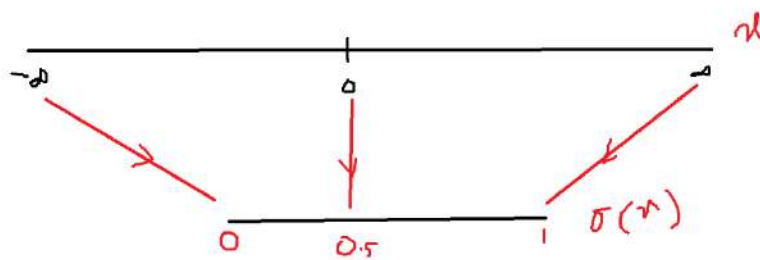
$$\sigma(0) = \frac{1}{1+e^{-(0)}}$$

$$\sigma(0) = \frac{1}{1+e^{-0}}$$

$$\sigma(0) = \frac{1}{1+1}$$

$$\sigma(0) = 0.5$$

x	$\sigma(x) = \frac{1}{1+e^{-x}}$
$-\inf$	0
0	0.5
$\inf$	1



$$p(\text{Subscription}) = \frac{1}{1+e^{-(b_0+b_1*age)}} = (3)$$

Problem in Equation - 3

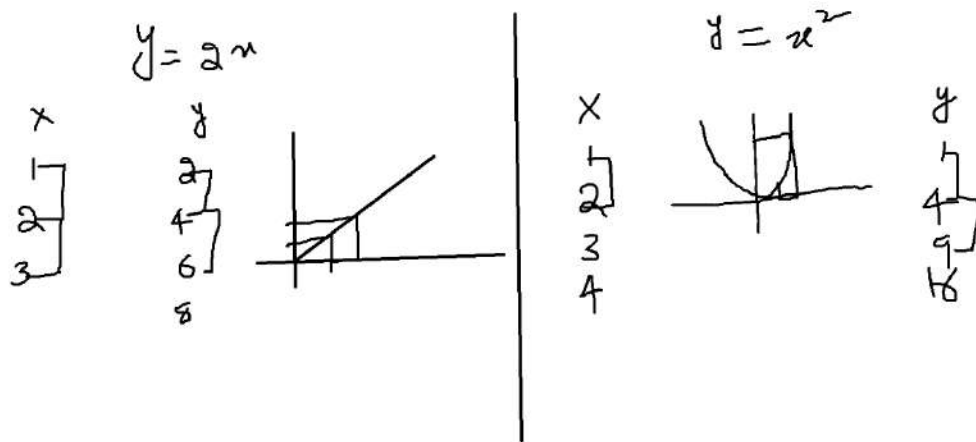
Interpretation of equation – 3

$$y = b_0 + b_1 * x_1$$

b_1 : If we increase x_1 by one unit then y is increasing by b_1 times

$$p(\text{Subscription}) = \frac{1}{1 + e^{-(b_0 + b_1 * \text{age})}} = (3)$$

Interpretation of b_1 from equation – 3



Interpretation is Not possible

- Because equation – 3 is a sigmoid function
- It looks like s curve
- Which is A non linear equation

Note: Only Linear equations we can interpret

Non linear equation we can not interpret

Solutions: Convert Non linear equation to Linear equation

$$p(y) = \frac{1}{1 + e^{-(b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n)}}$$

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n$$

$$p(y) = \frac{1}{1 + e^{-y}}$$

$$p(y)[1 + e^{-y}] = 1$$

$$p(y) + p(y) * e^{-y} = 1$$

$$p(y) * e^{-y} = 1 - p(y)$$

$$e^{-y} = \frac{1 - p(y)}{p(y)}$$

$$e^y = \frac{p(y)}{1 - p(y)}$$

$$\begin{array}{l} e^y = u \\ y = \log_e u \end{array}$$

$$\log_{10} a = 2$$

$$a = 10^2$$

$$\log_2 a = 4$$

$$a = 2^4$$

$$\log_c a = 2 \quad a = e^2$$

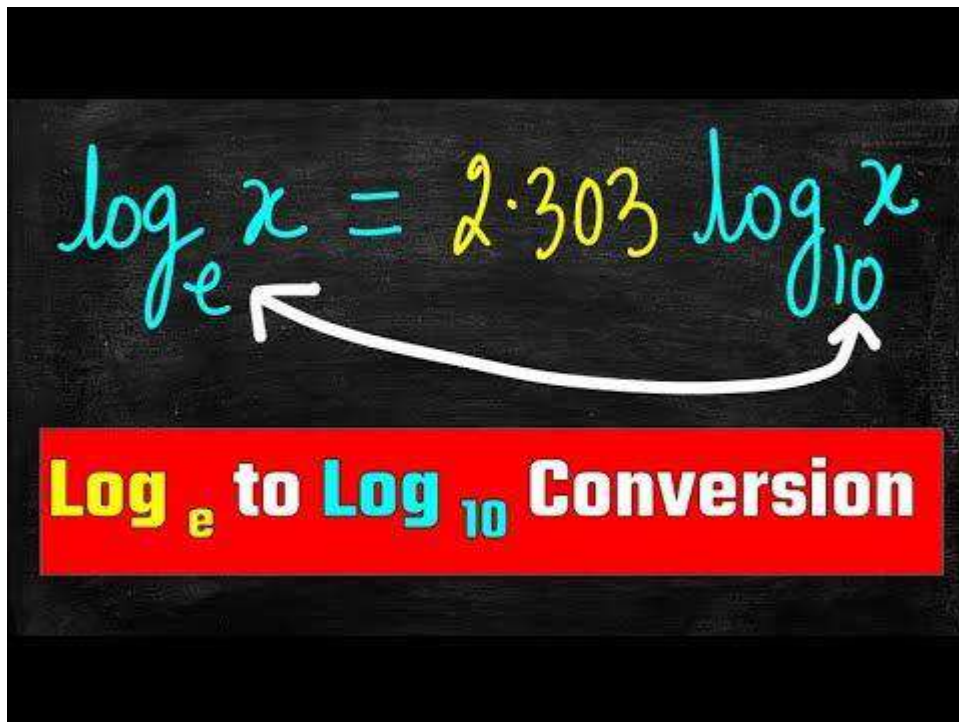
$$e^y = \frac{p(y)}{1 - p(y)}$$

$$y = \log_e \left(\frac{p(y)}{1 - p(y)} \right)$$

$$\log_e \left(\frac{p(y)}{1 - p(y)} \right) = y$$

$$\log_e \left(\frac{p(y)}{1 - p(y)} \right) = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n === (4)$$

$$\log_e \left(\frac{p(y)}{1 - p(y)} \right) = \ln \left(\frac{p(y)}{1 - p(y)} \right) = 2.303 * \log_{10} \left(\frac{p(y)}{1 - p(y)} \right)$$



$\text{Subscription} = b_0 + b_1 * \text{age} === (1)$ [Draw back : It is a regression but we want class]

$p(\text{Subscription}) = b_0 + b_1 * \text{age} == (2)$ [RHS = $-\infty$ to ∞ we want only 0 to 1]

$p(\text{Subscription}) = \frac{1}{1 + e^{-(b_0 + b_1 * \text{age})}} == (3)$ [Non linearity]

$$\log_e \left(\frac{p(y)}{1 - p(y)} \right) = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n === (4)$$

Interpretation:

$$y = b_0 + b_1 * x_1$$

if we increase x_1 by one unit , y will increase by b_1 times

$$\log_e \left(\frac{p(y)}{1 - p(y)} \right) = b_0 + b_1 * x_1$$

if we increase x_1 by one unit , $\log_e \left(\frac{p(y)}{1 - p(y)} \right)$ will increase by b_1 times

$$\log_e \left(\frac{p(y)}{1 - p(y)} \right) = \log \text{ odds ratio}$$

log odds ratio:

odds means different different

p(y) and 1 - p(y)

p(y) = yes 1 - p(y) = NO

p(y) = 0.6 yes

1 - p(y) = 0.4 no

$$\frac{p(y)}{1 - p(y)} = \text{odds ratio}$$

$$\frac{p(\text{yes})}{p(\text{no})} = \text{odds ratio}$$

$$\log_e \left(\frac{p(y)}{1 - p(y)} \right) = \log \text{ odds ratio}$$

Sir suppose we have three unique labels then how the ratio will differ

Rich poor middle

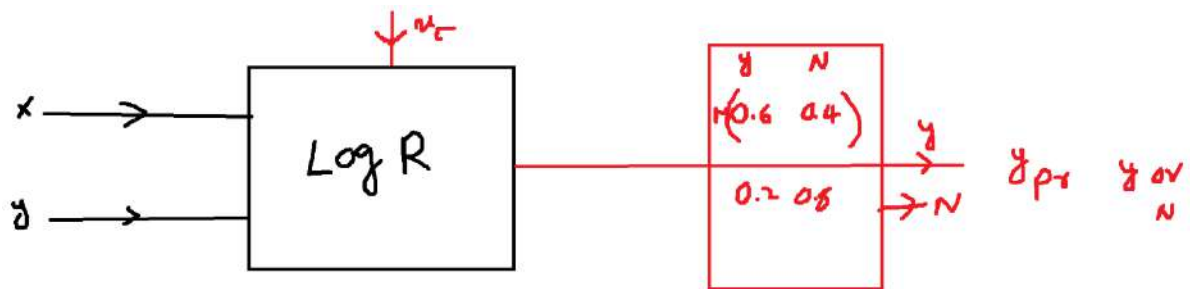
Rich vs poor+middle

if we increase x_1 by one unit, log odds ratio will increase by b_1 time

- 1) What is the meaning of classification problem
- 2) we achieved a logistic regression equation
- 3) sigmoid equation = $\frac{1}{1+e^{-x}}$ (activation function)
- 4) $p(y) = \frac{1}{1+e^{-(b_0+b_1*x_1)}}$ (here only we can get yes – no answer)
- 5) $\log_e \left(\frac{p(y)}{1-p(y)} \right) = b_0 + b_1 * x_1$ (This will give the interpretation)
- 6) log odds ratio

$$p(y = \text{yes}) = \frac{1}{1 + e^{-(b_0+b_1*x_1)}}$$

$$p(y = \text{no}) = \frac{1}{1 + e^{-(b_0+b_1*x_1)}}$$



Suppose a data set has 3 output labels

Output column name: zone

Labels: Red, yellow, green

whenever you pass the test data model will give 3 probabilities

$P(\text{zone} = \text{RED}), p(\text{zone} = \text{Yellow}), p(\text{zone} = \text{Green})$

$P(\text{zone} = \text{RED}) = 0.7$

$P(\text{zone} = \text{Yellow}) = 0.2$

$P(\text{zone} = \text{Green}) = 0.1$

maximum probability $p(\text{zone} = \text{Red}) = 0.7$, model

will give directly answer : RED zone

Step – 1: Inputs multiply with Coeff = $b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n$

Step – 2: $\frac{1}{1+e^{-(b_0+b_1*x_1+b_2*x_2+\dots+b_n*x_n)}}$ = belongs to 0 to 1 = probability

Step – 3: Number of classes = Number of probability values

Step – 4: which is the maximum probability final class

Loss function in Logistic regression

Cross Entropy or log loss

actual output = y_a predicted is = y_p predicted probability = p

$$- [y_a * \log(p) + (1 - y_a) * \log(1 - p)]$$

y_a	p	y_p	cross entropy	
0	$p(\text{yes}) = 0.9$	1	1	wrong prediction
1	0.9	1	0.045	correct prediction

$$- [y_a * \log(p) + (1 - y_a) * \log(1 - p)]$$

$$- [0 * \log(0.9) + (1 - 0) * \log(1 - 0.9)]$$

$$- (1) * \log(1 - 0.9) = 1$$

$$- [1 * \log(p) + (1 - 1) * \log(1 - p)]$$

$$- [1 * \log(0.9) + (0) * \log(1 - 0.9)]$$

$$- \log 0.9 = 0.045$$

Metrics are same for Regression

Metrics are same for classification

Metrics are same for cluster problems

Then why we are learning this math sir?

We need to learn ==== we need phd

How the models are developed ==== create our own models(We are not doing)

Why we are learning == ChatGPT

may be project timeline is big factor not to develop new algo

Rani

Sir will we do comparison between writing Linear Regression algorithm from scratch and using sklearn which one performs better ?

Distance between two points =